

Clinical Risk Assessment Report: Patient 49

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Positive for Metastasis
- Prediction Confidence: High Confidence
- Absolute Predicted Risk: 58.2% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 2.63x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as High-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET), which elevated the patient's absolute risk by +9.2%. Biologically, this feature points to continuous, elongated bands of high-density or highly active tumor tissue.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: 1.39 [Top 8%] (+9.2%)**
Points to continuous, elongated bands of high-density or highly active tumor tissue.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.41 [Bottom 2%] (+5.5%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

25th Percentile Metabolic Activity (CONVENTIONAL_SUVbwQ1) **Value: 1.34 [Top 9%] (+4.3%)**
Reflects the baseline metabolic activity and glucose consumption of the tumor tissue.

Patient Age (Age) **Value: 30.0 [Bottom <1%] (+3.9%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 0.8 [Top 20%] (+3.5%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Short Run High Gray-Level Emphasis (CT) (GLRLM_SRHGE.CT) **Value: 0.74 [Avg (73th %ile)] (+3.4%)**
Reflects fragmented, short streaks of high density or metabolism, indicating a disjointed and aggressive tumor architecture.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.28 [Bottom <1%] (+3.3%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: 4.1 [Top <1%] (+3.2%)**
Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

3. Protective / Mitigating Factors (Reducing Risk)

No major risk-reducing features observed in the top drivers.

Machine Learning Feature Attribution (SHAP Decision Path)

